

Constructive Master's Thesis Work in Industry

Guidelines for Applying Design Science Research
(guest lecture DAT246/DIT246 Empirical SE)

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October 10, 2022

Big shout-out to **Jennifer Horkoff**, these slides are heavily inspired by our joint DSR lecture on Bachelor level.

Theses we have worked on...

- ▶ Feedback system (integrated in test car) to facilitate requirements feedback flows in iterative development
- ▶ Concept and IDE plugin to improve unit testing practices with gamification
- ▶ Automation of validating non-functional test results
- ▶ Testing strategies to support continuous integration for complex systems
- ▶ Low cost construction of coordination and communication networks from systems engineering data
- ▶ Test automation strategy to enable continuous integration for an automotive platform
- ▶ Requirements elicitation approach for intelligent and interactive automotive systems
- ▶ Architecture framework for blockchain implementation
- ▶ Concept and tool support for managing textual requirements in git
- ▶ Requirements engineering approach for developing cloud support of autonomous trucks
- ▶ Flexible integration concept of voice recognition in an automotive android platform

Outline

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Conclusion

What to expect...

- ▶ Experiences from 12 master's theses supervised over the last seven years.
- ▶ Presented as a design science research report. [Access and cite as \[Knauss, 2021\]](#)

The artifact

- ▶ *Guidelines to conduct a design science master's thesis in industry*
- ▶ Aim: complement existing
 - ▶ method descriptions
[Vaishnavi and Kuechler, 2007, Engström et al., 2020]
 - ▶ guidelines [Hevner et al., 2004]
 - ▶ practices
[Wieringa, 2009, Storey et al., 2017]

The knowledge questions

- RQ1 Which problems are frequently encountered when conducting a master's thesis in industry based on design science research methodology?
- RQ2 Which potential solutions exist to mitigate problems with running a design science master's thesis?
- RQ3 To what extent can typical problems with running a design science master's thesis be addressed?

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Motivation

Some quotes

"Industry wants us to program, not to work on concepts only."

–Student to academic supervisor

"University does not allow us to implement stuff, they want us to work theoretically."

–Student to industry supervisor

"???"

–Dialogue of industry and academic supervisor

Some Alternatives

- ▶ Build the artifact and run experiment
- ▶ Build no artifact and do exploratory qualitative case study
- ▶ Action research: Emphasise intervention, not design
- ▶ Take a summer job

Stakeholders

Motivation

Who?	Student	Industry supervisor	Academic supervisor	Examiner
What?	High quality education and impact on CV.	Learn about topic, screen potential employees, solve problem.	Learn about topic, contribution to research, contact to industry.	High quality thesis
Sentiment?	Positive	Positive	Positive	Indifferent
Power?	Walk away from topic, fine-tune where focus is put	Walk away from topic or future theses, manage resources	Competence in topic area	Pass/fail decision

Sciences of the Artificial [Simon, 1996]

- ▶ “Historically and traditionally, it has been the task of the science disciplines to teach about
natural things: how they are and how they work.
- ▶ It has been the task of engineering schools to teach about
artificial things:
how to make artifacts that have desired properties and how to design.
- ▶ Engineering, medicine, business, architecture, and painting are concerned not with the necessary but with the contingent
– not with how things are but how they might be. . .”

Categories of scientific disciplines [van Aken, 2004]

Formal sciences

- ▶ E.g., philosophy and mathematics
- ▶ Empirically “void”, main test internal logical consistency

Explanatory sciences

- ▶ E.g., natural sciences and major section of social sciences
- ▶ Describe, explain and possibly predict observable phenomena within its field

Design sciences

- ▶ Engineering sciences, medical sciences and modern psychotherapy
- ▶ Solve construction problems or improvement problems

Design Science Definition

The design-science paradigm seeks to extend the boundaries of human and organizational capabilities by creating new and innovative **artifacts**

[[Hevner et al., 2004](#)]



"Sir Isaac Newton, renowned inventor of ... the catflap!"

"The what?" said Richard.

"The catflap! A device of the utmost cunning, perspicuity and invention. It is a door within a door, you see, a ..."

"Yes," said Richard, **"there was also the small matter of gravity."**

"Gravity,"
said Dirk with a slightly dismissed shrug, "yes, there was that as well, I suppose. Though that, of course, was merely a discovery. It was there to be discovered." ...

"You see?" he said dropping his cigarette butt,
"They even keep it on at weekends. Someone was bound to notice sooner or later."

But the catflap ... ah, there is a very different matter. Invention, pure creative invention. It is a door within a door, you see."

Slide via
Tomi
Männistö

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State of the art

Design Science

Three types of research questions for DSR, relating to *the way of knowing, the way of framing, and the way of designing* [Thuan et al., 2019].

Must reads

- ▶ Method descriptions [Vaishnavi and Kuechler, 2007, Engström et al., 2020]
- ▶ Guidelines [Hevner et al., 2004]
- ▶ Practices: Regulative cycle [Wieringa, 2009], graphical abstract [Storey et al., 2017]

A key part of the graphical abstract [Storey et al., 2017] is the technology rule [Aken, 2005]:
"To achieve <Effect> in <Situation> apply <Intervention>"

State of the art

Design Science

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- Answer question 1

Three types of research questions for DSR, relating to *the way of knowing, the way of framing, and the way of designing* [Thuan et al., 2019].

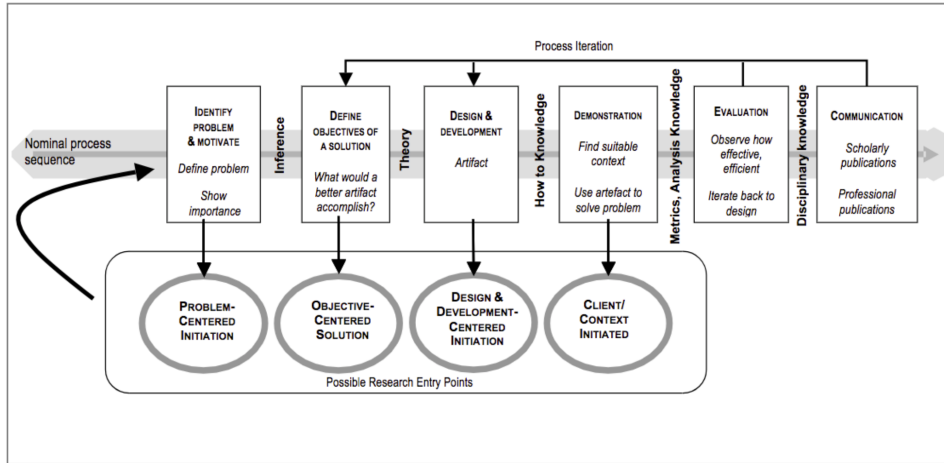
Must reads

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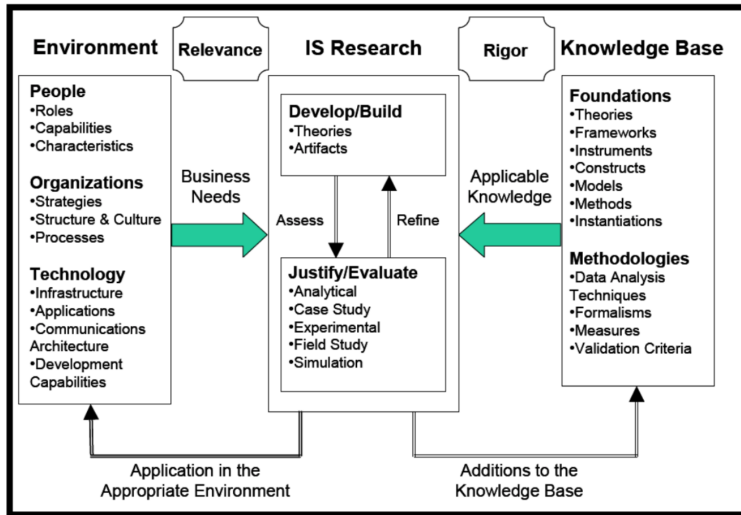
A key part of the graphical abstract [Storey et al., 2017] is the technology rule [Aken, 2005]:
“To achieve <Effect> in <Situation> apply <Intervention>”

The Design Science Research Methodology

[Peffers et al., 2007]



Design Science as per [Hevner et al., 2004]



Design Science Guidelines [Hevner et al., 2004]

Guideline	Description
Guideline 1: Design as an Artifact	Design-science research must produce a viable artifact in the form of a construct, a model, a method, or an instantiation.
Guideline 2: Problem Relevance	The objective of design-science research is to develop technology-based solutions to important and relevant business problems.
Guideline 3: Design Evaluation	The utility, quality, and efficacy of a design artifact must be rigorously demonstrated via well-executed evaluation methods.
Guideline 4: Research Contributions	Effective design-science research must provide clear and verifiable contributions in the areas of the design artifact, design foundations, and/or design methodologies.
Guideline 5: Research Rigor	Design-science research relies upon the application of rigorous methods in both the construction and evaluation of the design artifact.
Guideline 6: Design as a Search Process	The search for an effective artifact requires utilizing available means to reach desired ends while satisfying laws in the problem environment.
Guideline 7: Communication of Research	Design-science research must be presented effectively both to technology-oriented as well as management-oriented audiences.

Hevner et al. 2004

Design Evaluation

IT artifacts can be evaluated in terms of functionality, completeness, consistency, accuracy, performance, reliability, usability, fit with the organization, and other relevant quality attributes.

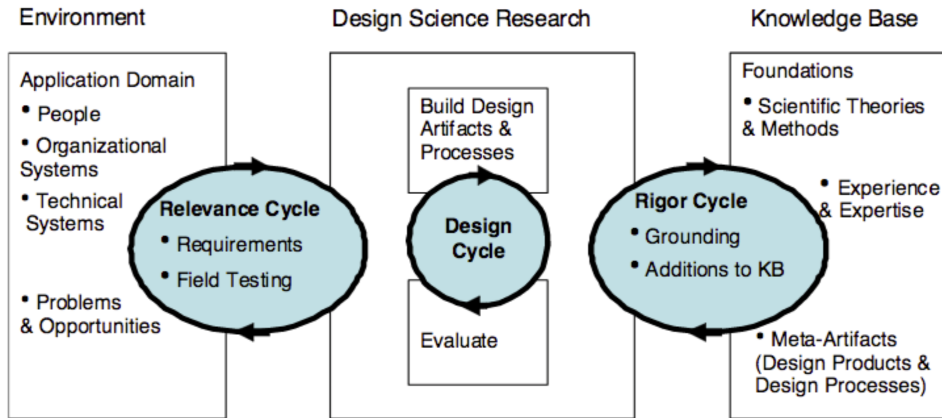
[[Hevner et al., 2004](#)]

Comment: Pragmatically one should evaluate whether it solves the problem.

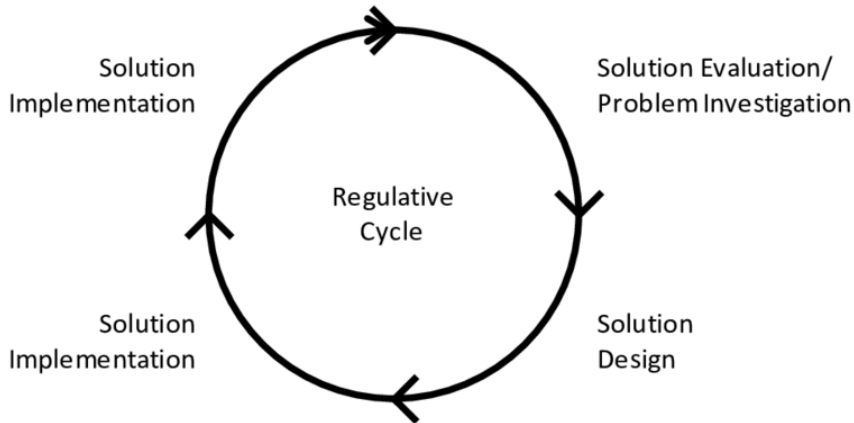
Design Evaluation Methods [Hevner et al., 2004]

1. Observational	Case Study: Study artifact in depth in business environment
	Field Study: Monitor use of artifact in multiple projects
2. Analytical	Static Analysis: Examine structure of artifact for static qualities (e.g., complexity)
	Architecture Analysis: Study fit of artifact into technical IS architecture
	Optimization: Demonstrate inherent optimal properties of artifact or provide optimality bounds on artifact behavior
	Dynamic Analysis: Study artifact in use for dynamic qualities (e.g., performance)
3. Experimental	Controlled Experiment: Study artifact in controlled environment for qualities (e.g., usability)
	Simulation – Execute artifact with artificial data
4. Testing	Functional (Black Box) Testing: Execute artifact interfaces to discover failures and identify defects
	Structural (White Box) Testing: Perform coverage testing of some metric (e.g., execution paths) in the artifact implementation
5. Descriptive	Informed Argument: Use information from the knowledge base (e.g., relevant research) to build a convincing argument for the artifact's utility
	Scenarios: Construct detailed scenarios around the artifact to demonstrate its utility

Design Science as per [Hevner, 2007]

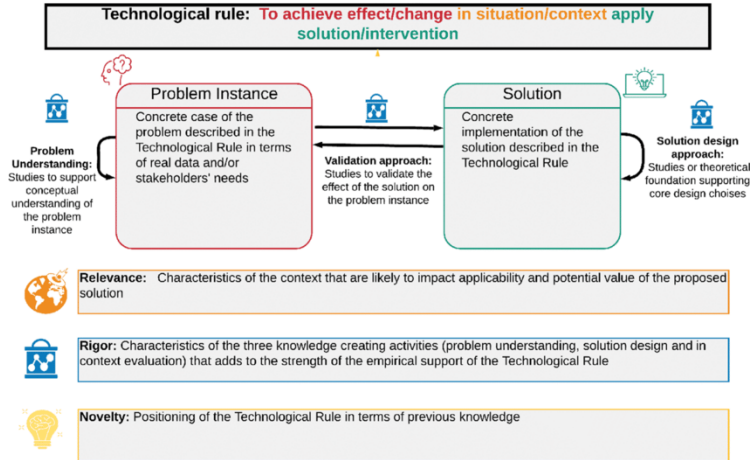


Regulative Cycle [Wieringa, 2009]



The graphical abstract template [Storey et al., 2017]

Generic Visual Abstract



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Method

Data collection

- ▶ Analysing notes and memories from the interaction between supervisor and student
 - ▶ Feedback from examiners and industry partners
 - ▶ Review comments from reviewers of publications derived from theses
 - ▶ Applying (improved) guidelines every year
 - ▶ Guest lecture on design science research in BSc program
 - ▶ Survey with faculty, grad students, and thesis students
- ▶ Guidelines hopefully valuable beyond scope of investigation
i.e. the two universities and programs
 - ▶ Yet: research method relates to this particular context

<i>Cycle</i>	<i>Data type</i>	<i>Artifact</i>	<i>Source</i>
2014	MT, CP	Feedback system (integrated in test car) to facilitate requirements feedback flows in iterative development	[Andersson and Rybacki, 2014, Knauss et al., 2015]
2015	MT	Concept and IDE plugin to improve unit testing practices with gamification	[Arnarsson and Jóhannesson, 2015] [Ismail and Fagerström, 2015, Fagerström et al., 2016]
2015	MT, CP	Automation of validating non-functional test results	
2016	MT	Testing strategies to support continuous integration for complex systems	[Gangineni and Jamil, 2016]
2016	MT, JP	Loco Coco: Low Cost construction of Coordination and Communication networks from model-based systems engineering data	[Mohamad, 2016, Mohamad et al., 2017]
2017	MT	Test automation strategy to enable continuous integration for an automotive platform	[Simon and Balabhadrapatruni, 2017]
2018	MT	Requirements elicitation approach for intelligent and interactive systems in autonomous vehicles	[Shahin and Curta, 2018]
2019	MT	Architecture Framework for Blockchain Implementation	[Sjöberg, 2019] [Knauss et al., 2018, Gebremichael, 2019, Alsahhar and Bulai, 2019]
2019	CP, MT, BT	Concept and tool-support for managing textual (system) requirements in git (T-Reqs)	
2020	MT	Requirements engineering approach for developing cloud based support of autonomous trucks	[Supriya, 2020]
2020	MT	Flexible integration concept of voice recognition components in an automotive android platform	[Johansson and Eliasson, 2020]

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The artifact: Guidelines

(G1) Define the **artifact** early. Use it to agree with all stakeholders on the thesis goals with respect to knowledge contribution.

The artifact: Guidelines

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- Answer Question 2

(G1) Define the **artifact** early. Use it to agree with all stakeholders on the thesis goals with respect to knowledge contribution.

The artifact: Guidelines

(G2) Work in **iterations**. Improve the artifact and the knowledge in each iteration. Specifically, contribute to each research question in each iteration.

(G3) Define **research questions** with respect to the regulative cycle [Hevner et al., 2004], i.e. one related to the problem, one related to potential solutions and their construction, and one related to evaluation (match solution to problem).

Example

RQ1 What is the problem with the current way of doing X?

RQ2 Which potential solutions could mitigate this problem?

RQ2.1 (optional) Which potential solutions are most promising with respect to potential value / effort of implementing?

RQ2.2 (optional) How can a suitable solution be implemented in the artifact?

RQ3 To what extent can the problem with the current way of doing X be solved by potential solutions?

(G3) Define **research questions** with respect to the regulative cycle [Hevner et al., 2004], i.e. one related to the problem, one related to potential solutions and their construction, and one related to evaluation (match solution to problem).

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- Answer Question 3

Example

- RQ1 What is the problem with the current way of doing X?
- RQ2 Which potential solutions could mitigate this problem?
 - RQ2.1 (optional) Which potential solutions are most promising with respect to potential value / effort of implementing?
 - RQ2.2 (optional) How can a suitable solution be implemented in the artifact?
- RQ3 To what extent can the problem with the current way of doing X be solved by potential solutions?

The artifact: Guidelines

(G4) Have regular **meetings**. Student and supervisor meet once per phase (once per week). Student, industry supervisor, and academic supervisor meet once per cycle (once per month). Goal: support rigor.

(G5) **Shift emphasis** between cycles. Work on each research question in each cycle, but put more emphasis on RQ1 (Problem) in Cycle one, on RQ2 (Solutions) in Cycle two, and on RQ3 (Evaluation) in Cycle three.

Cycle	<i>Artifact Impl.</i>	RQ 1 (Problem) <i>Problem investigation</i>	RQ 2 (Solutions) Solution design, design validation	RQ 3 (Evaluation) Evaluation
I	Incremental work	n interviews, ...	Lit. review brainstorming, ...	Prototype, usability test, interviews, brief survey, ...
II	Incremental work	n interviews (merge with evaluation interviews in Cycle I)	Design Workshop, ...	Observe practitioners using prototype, interviews, ...
III	Incremental work	n interviews (merge with evaluation interviews in Cycle II)	Design Workshop, ...	Practitioners use artifact to solve problem, life demo...

The artifact: Guidelines

(G6) Have a dedicated section to **describe the artifact**. A concise description of the artifact is beneficial to the casual reader and allows other sections (e.g. findings, discussion) to focus on the learning, not the artifact.

The artifact: Guidelines

(G7) **Write the thesis** document as you go, but do not submit it that way. Often, the best way to report your results can only be decided after Cycle 3. Thus, chose the least obstructive way of documenting the results during the thesis (chronological). Then restructure before submission.

3.2 Solution candidates
3.3 Validation
3.4 Implementation
3.5 Evaluation

3.2 Solution candidates
3.3 Validation
3.4 Implementation
3.5 Evaluation

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4 Findings
4.1 Cycle I Findings
4.1.1 Problem
4.1.2 Solution candidates
4.1.3 Validation
4.1.4 Implementation
4.1.5 Evaluation
4.2 Cycle II Findings
4.2.1 Problem
4.2.2 Solution candidates
4.2.3 Validation
4.2.4 Implementation
4.2.5 Evaluation
4.3 Cycle III Findings
4.3.1 Problem
4.3.2 Solution candidates
4.3.3 Validation
4.3.4 Implementation
4.3.5 Evaluation

4. The Artifact

5. The Artifact

5. Findings
5.1 RQ1 (Problem)
5.2 RQ2 (Solution candidate)
5.3 RQ3 (Evaluation)

6. Discussion
6.1 Answer to RQ 1 (Problem)
6.2 Answer to RQ 2 (Solution)
6.3 Answer to RQ 3

6. Discussion
6.1 Implications for research
6.2 Implications for practice

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Results

In the paper

- ▶ Per year description of what was learned
- ▶ RQ1: Underlying problems that must be addressed.
- ▶ RQ2: Basically the guidelines, but also those that were discarded.
- ▶ RQ3: Some evaluation

RQ1: Which problems are frequently encountered when conducting a master's thesis in industry based on design science research methodology?

- ▶ Balance: constructive work, practical relevance, academic relevance
- ▶ Practical problem often initially not well understood
 - ▶ By all stakeholders (student and academic/industry supervisor)!
 - ▶ Solving it in a real world environment: **very hard problem**.
- ▶ Difficult to report iterative work in thesis
- ▶ Difficult to leverage different stakeholder perspectives
- ▶ Hard to plan iterative research cycles and to formulate research questions

RQ2 Which potential solutions exist to mitigate problems with running a design science master's thesis?

Beyond my guidelines...

- ▶ Guidelines by [Hevner et al., 2004], regulative cycle [Wieringa, 2009], graphical abstract [Storey et al., 2017, Engström et al., 2020] (use them!)
- ▶ 4th cycle to generalize artifact beyond case company (high risk, low gain?)
- ▶ Including examiner in deciding research plan and trade-offs (potential gain, but defeats separation of roles)

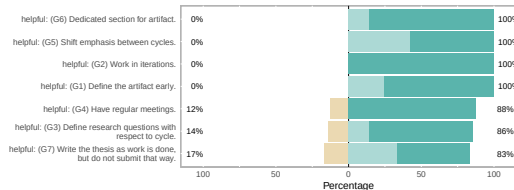
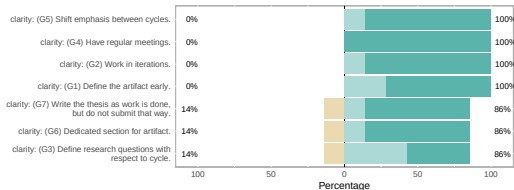
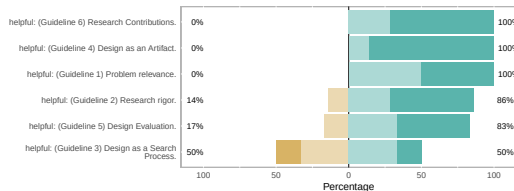
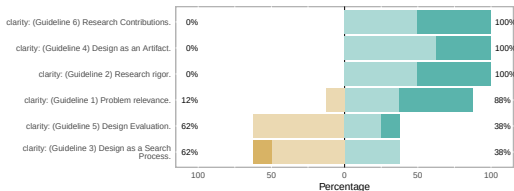
RQ3 To what extent can typical problems with running a design science master's thesis be addressed?

PhD student

"I do not get the activities I should perform from these guidelines [referring to Guidelines by Hevner et al.]. I however think they are helpful in checking that one is not below required design science standards but not in doing the research itself."

- ▶ Positive: Target students, help getting started
- ▶ Negative: Some participants worry that guidelines are too specific (i.e. G3) or too general (G4, G7)

RQ3 To what extent can typical problems with running a design science master's thesis be addressed?



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Design Research – Theory Development

- ▶ Researcher proceeds through **iterative cycles** of design and implementation
- ▶ Use each implementation as an opportunity to collect data to inform subsequent design
- ▶ Parallel and retrospective process of reflection upon the design and its outcomes
 - ▶ Elaborate upon the initial hypotheses and principles,
 - ▶ Refining, adding, and discarding
 - ▶ Gradually putting together a coherent theory that reflects the understanding of the design experience

Theory Types in Information Systems [Gregor, 2006]

Table 2. A Taxonomy of Theory Types in Information Systems Research	
Theory Type	Distinguishing Attributes
I. Analysis	Says what is. The theory does not extend beyond analysis and description. No causal relationships among phenomena are specified and no predictions are made.
II. Explanation	Says what is, how, why, when, and where. The theory provides explanations but does not aim to predict with any precision. There are no testable propositions.
III. Prediction	Says what is and what will be. The theory provides predictions and has testable propositions but does not have well-developed justificatory causal explanations.
IV. Explanation and prediction (EP)	Says what is, how, why, when, where, and what will be. Provides predictions and has both testable propositions and causal explanations.
V. Design and action	Says how to do something. The theory gives explicit prescriptions (e.g., methods, techniques, principles of form and function) for constructing an artifact.

Advice for students

Conclusion

- ▶ Fully use the supervision team: both academic and industry supervisors!
- ▶ Use them as sounding board in each phase and when planning your next cycle.
- ▶ Provide value to practitioners early, thus they are more likely to participate in your research.

Advice for students

Conclusion

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- ▶ Fully use the supervision team: both academic and industry supervisors!
- ▶ Use them as sounding board in each phase and when planning your next cycle.
- ▶ Provide value to practitioners early, thus they are more likely to participate in your research.

Advice for practitioners

Conclusion

The more active you are, the more you will get out of thesis work!

- ▶ Help to clarify the actual problem, it is often much more complicated and hard to understand than you may believe.
- ▶ Provide access to different internal stakeholders and moderate the process.
- ▶ Help the student to discuss the practical relevance of their work, and in particular make sure that the artifact is of value to you and your organization.

Advice for supervisors

Conclusion

- ▶ Hands-on!
- ▶ Tough problem to solve, a lot can go wrong.
- ▶ Mind the lack in constructive alignment with respect to constructive theses!

Advice for examiners

Conclusion

- ▶ Hard to compare DSR to other empirical methods (e.g. in terms of data)
- ▶ Interviews are often more structured and targeted, but less formal
- ▶ Analysis is often more straight forward (constrained by artifact)
- ▶ Total number of interviewees might be lower (since they might get asked several times)
- ▶ Iterations help to increase quality
- ▶ Artifact provides deeper insights
- ▶ Consider effort for managing iterations and building artifact

Open questions

Conclusion

- ▶ What if the problem is not sufficiently solved after the thesis?
 - ▶ Need to report the thesis as something complete
 - ▶ Awkward to start the next thesis where the first left of: does it stand on its own?
- ▶ How to formalize problems that are initially too vague?

Open questions

Conclusion

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- Answer Question 5

- ▶ What if the problem is not sufficiently solved after the thesis?
 - ▶ Need to report the thesis as something complete
 - ▶ Awkward to start the next thesis where the first left of: does it stand on its own?
- ▶ How to formalize problems that are initially too vague?

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